

Project 4

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Assuming the single index model

$$R_i = \alpha_i + \beta_i R_m + \epsilon_i, \quad \text{cov}(\epsilon_i, R_m) = 0, \quad \text{cov}(\epsilon_i, \epsilon_j) = 0 \quad (i \neq j)$$

over 01-Jan-2017 to 01-Jan-2022. This is the same window as Projects 1 and 2, so the return series is the same 59 months on the same 30 stocks, with $\hat{R}_{GSPC}$ as the index R_m . Part 3 has to be drawn on Project 2's axes, so using anything else would make the two frontiers incomparable.

1. Estimating $\alpha_i, \beta_i, \sigma_{\epsilon_i}^2$

Each stock's return is regressed on the index by ordinary least squares. $\sigma_{\epsilon_i}^2$ is the residual variance, $\text{RSS}/(T-2)$ with $T = 59$.

Table 1: Single index model estimates, monthly, 30 stocks

Stock	alpha (%)	beta	sigma^2_e	R^2
NVDA	3.1289	1.3118	0.013002	0.2128
AAPL	1.8708	1.2019	0.004528	0.3945
MSFT	1.9232	0.8625	0.001318	0.5355
AVGO	1.3326	0.9597	0.003649	0.3402
AMD	3.2349	1.8741	0.019208	0.2719
NOW	2.2970	1.0001	0.003677	0.3572
GOOG	0.9783	1.0576	0.002004	0.5327
META	0.1958	1.2828	0.003509	0.4892
NFLX	1.8167	0.8136	0.008091	0.1432
VZ	-0.0475	0.4422	0.001532	0.2068
DIS	-0.6524	1.1871	0.003869	0.4266
T	-1.1041	0.7345	0.001829	0.3760
V	0.5720	0.9692	0.001957	0.4950
JPM	-0.0061	1.1351	0.002338	0.5295
MS	-0.1021	1.5381	0.002467	0.6620
GS	-0.6535	1.4983	0.002851	0.6166
BAC	-0.3476	1.4827	0.002393	0.6524
SCHW	0.2161	1.0448	0.004477	0.3324
AMZN	1.2181	1.0959	0.004131	0.3725
TSLA	4.2330	1.9779	0.031628	0.2017

Stock	alpha (%)	beta	sigma^2_e	R^2
MCD	0.8550	0.5982	0.001804	0.2883
SBUX	0.5479	0.8235	0.003175	0.3037
HD	0.9475	0.9825	0.002057	0.4894
EBAY	0.1617	1.1172	0.005145	0.3313
LLY	2.1275	0.3633	0.004996	0.0512
NVO	1.6933	0.4391	0.002770	0.1245
UNH	1.1028	0.8612	0.002903	0.3429
JNJ	0.0470	0.7387	0.001442	0.4360
AMGN	0.2093	0.6394	0.003714	0.1835
PFE	0.7564	0.6977	0.003999	0.1991

```
#> market:  mean 1.3579%   sd 4.4802%
#> beta   :  0.363 to 1.978, mean 1.024
#> R^2    :  0.051 to 0.662, mean 0.363
```

The single index model changes the covariance matrix only. The mean vector feeding the frontier is untouched, so any difference in part 3 comes from Σ alone.

2. The single index variance-covariance matrix

$$\sigma_{ij} = \beta_i \beta_j \sigma_m^2 \quad (i \neq j), \quad \sigma_{ii} = \beta_i^2 \sigma_m^2 + \sigma_{\epsilon_i}^2$$

so in matrix form $\Sigma_{SIM} = \sigma_m^2 \beta \beta' + \text{diag}(\sigma_{\epsilon}^2)$.

```
#> dimension           30 x 30, symmetric
#> smallest eigenvalue  1.346e-03 (positive definite)
#> condition number     60.9   vs 831.4 for the historical matrix
#> free parameters      61     vs 465, from 59 months of data
```

Table 2: Top-left corner of the single index matrix

	NVDA	AAPL	MSFT	AVGO	AMD	NOW
NVDA	0.01650	0.00316	0.00227	0.00253	0.00493	0.00263
AAPL	0.00316	0.00743	0.00208	0.00232	0.00452	0.00241
MSFT	0.00227	0.00208	0.00281	0.00166	0.00324	0.00173
AVGO	0.00253	0.00232	0.00166	0.00550	0.00361	0.00193
AMD	0.00493	0.00452	0.00324	0.00361	0.02630	0.00376
NOW	0.00263	0.00241	0.00173	0.00193	0.00376	0.00568

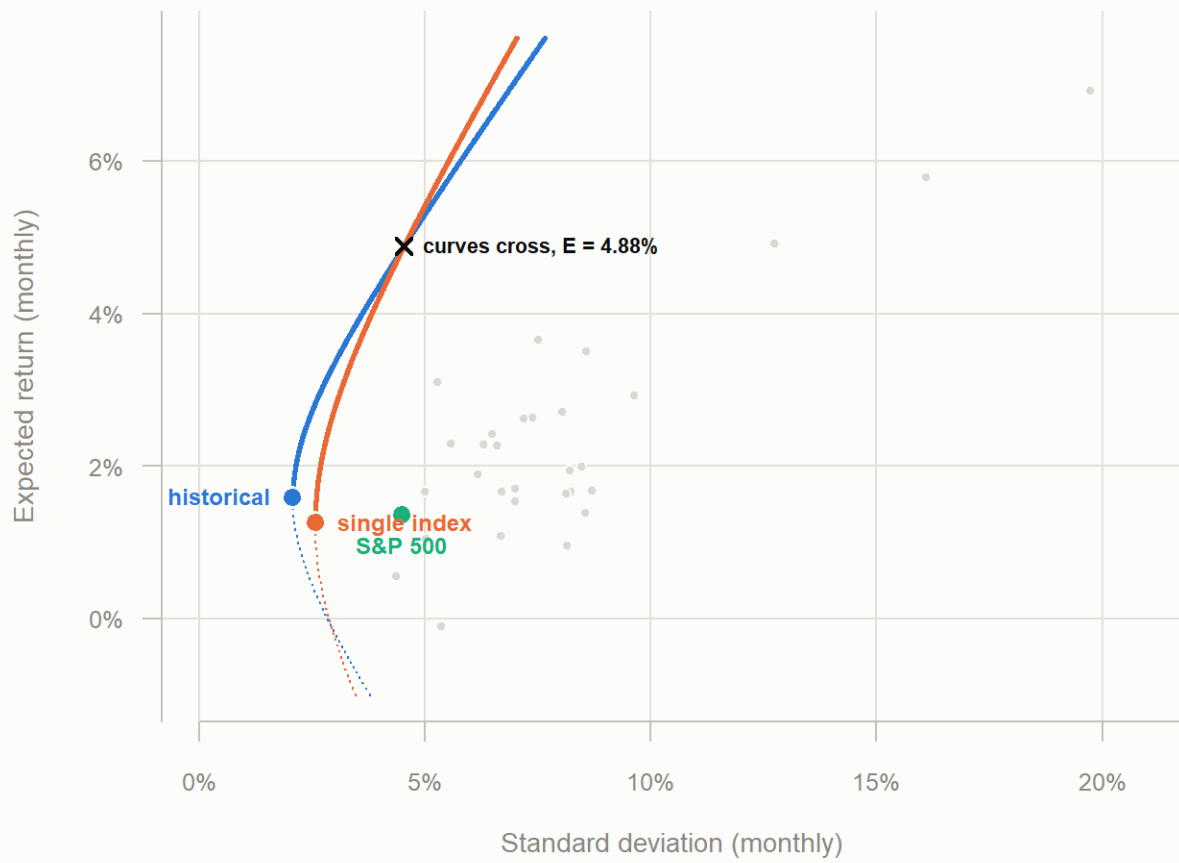
3. The frontier under both matrices

Repeating Project 2 part (e) — the hyperbola $\sigma^2 = 1/C + (C/D)(E - A/C)^2$ — with Σ_{SIM} in place of the historical matrix, holding $\bar{R}$ fixed.

Matrix	A	B	C	D	E min (%)	sd min (%)
Historical (Project 2)	36.9864	1.25629	2330.03	1559.19	1.587	2.072
Single index	19.0688	1.18367	1518.53	1433.81	1.256	2.566

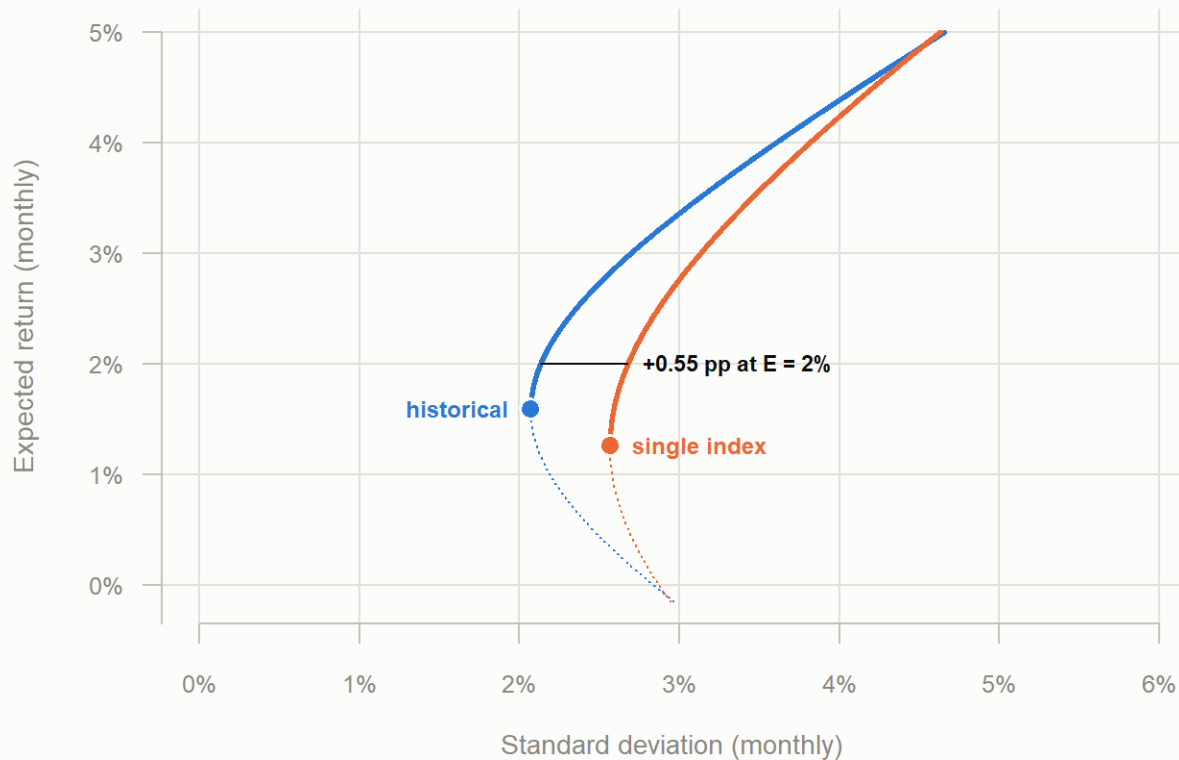
Two frontiers: same means, different covariance matrices

Solid: efficient half. Dotted: inefficient half. Grey: the 30 stocks.



The same two frontiers, near the vertex

Over the range the data occupies, the single index frontier sits to the right.



Appendix: full code

```
# Project 4, parts 1-3: the single index model
#
#  $R_i = \alpha_i + \beta_i R_m + \epsilon_i$ 

local({
  f <- sub("^--file=", "", grep("^--file=", commandArgs(FALSE), value = TRUE))
  if (length(f) == 1L && basename(f) == "single_index.R")
    setwd(dirname(normalizePath(f)))
})

DATA <- ".."

# ---- data -----

a <- read.csv(file.path(DATA, "stockData_train.csv"), sep = ",", header = TRUE,
              check.names = FALSE)
stopifnot(!any(duplicated(names(a))))

prices <- a[, -1]
dates <- as.Date(prices$Date)
P <- as.matrix(prices[, -1])
stopifnot(nrow(P) == 60, ncol(P) == 31, !anyNA(P))
```

```

n <- nrow(P)
ret <- P[-1, ] / P[-n, ] - 1 # 59 x 31

stk <- setdiff(colnames(ret), "^GSPC")
ret30 <- ret[, stk]
Rm <- as.numeric(ret[, "^GSPC"]) # the index, our single factor
Rbar <- colMeans(ret30)
S_hist <- cov(ret30) # Project 2's historical matrix

Tn <- nrow(ret30); N <- ncol(ret30)
stopifnot(Tn == 59, N == 30)

# ---- 1. regress each stock on the S&P 500 -----

fit <- lapply(stk, function(s) lm(ret30[, s] ~ Rm))
names(fit) <- stk
alpha <- vapply(fit, function(f) unname(coef(f)[1]), numeric(1))
beta <- vapply(fit, function(f) unname(coef(f)[2]), numeric(1))
sig2e <- vapply(fit, function(f) summary(f)$sigma^2, numeric(1)) # RSS/(T-2)
r2 <- vapply(fit, function(f) summary(f)$r.squared, numeric(1))

stopifnot(max(abs(beta - apply(ret30, 2, function(x) cov(x, Rm) / var(Rm)))) < 1e-12)
stopifnot(max(abs(alpha - (Rbar - beta * mean(Rm)))) < 1e-15)
stopifnot(max(abs(sig2e / vapply(fit, function(f) sum(residuals(f)^2) / (Tn - 2),
                                numeric(1)) - 1)) < 1e-14)

stopifnot(max(abs(alpha + beta * mean(Rm) - Rbar)) < 1e-16)

cat("1. single index model estimates (monthly)\n")
cat(sprintf(" market mean %.4f%%, market sd %.4f%%\n",
            mean(Rm) * 100, sd(Rm) * 100))
cat(sprintf(" beta range %.3f to %.3f (mean %.3f)\n",
            min(beta), max(beta), mean(beta)))
cat(sprintf(" alpha range %.4f%% to %.4f%% per month\n",
            min(alpha) * 100, max(alpha) * 100))
cat(sprintf(" R^2 range %.3f to %.3f (mean %.3f)\n\n",
            min(r2), max(r2), mean(r2)))

sim_tbl <- data.frame(alpha = alpha, beta = beta, sigma2_e = sig2e, R2 = r2)

# ---- 2. the single index variance-covariance matrix -----

sig2m <- var(Rm)
S_sim <- outer(beta, beta) * sig2m
diag(S_sim) <- beta^2 * sig2m + sig2e
dimnames(S_sim) <- list(stk, stk)

stopifnot(isSymmetric(S_sim))
stopifnot(all.equal(diag(S_sim), beta^2 * sig2m + sig2e))

ev_sim <- eigen(S_sim, only.values = TRUE)$values
ev_hist <- eigen(S_hist, only.values = TRUE)$values
stopifnot(min(ev_sim) > 0)

```

```

cat("2. variance-covariance matrix from the single index model\n")
cat(sprintf("  dim %d x %d, symmetric, smallest eigenvalue %.3e\n",
  nrow(S_sim), ncol(S_sim), min(ev_sim)))
cat(sprintf("  condition number: single index %.1f  historical %.1f\n",
  max(ev_sim) / min(ev_sim), max(ev_hist) / min(ev_hist)))
cat(sprintf("  free parameters: single index %d, historical %d, from %d months\n",
  2 * N + 1, N * (N + 1) / 2, Tn))
cat(sprintf("  mean |off-diagonal|: single index %.6f, historical %.6f\n\n",
  mean(abs(S_sim[upper.tri(S_sim)])),
  mean(abs(S_hist[upper.tri(S_hist)]))))

# ---- 3. frontier from each matrix -----

frontier <- function(Sig, mu) {
  Si <- solve(Sig); one <- rep(1, length(mu))
  A <- as.numeric(t(one) %*% Si %*% mu)
  B <- as.numeric(t(mu) %*% Si %*% mu)
  C <- as.numeric(t(one) %*% Si %*% one)
  D <- B * C - A^2
  stopifnot(B > 0, C > 0, D > 0)
  list(A = A, B = B, C = C, D = D, Emin = A / C, sdmin = sqrt(1 / C),
    xmin = as.numeric(Si %*% one / C))
}
f_hist <- frontier(S_hist, Rbar)
f_sim <- frontier(S_sim, Rbar)

# The historical numbers must reproduce Project 2 part (a) exactly
stopifnot(abs(f_hist$A - 36.9864) < 1e-3, abs(f_hist$B - 1.25629) < 1e-4,
  abs(f_hist$C - 2330.03) < 1e-1, abs(f_hist$D - 1559.19) < 1e-1)

sd_at <- function(f, E) sqrt(1 / f$C + (f$C / f$D) * (E - f$A / f$C)^2)

# The hyperbola and parabola forms must agree, for both matrices
for (fr in list(f_hist, f_sim)) {
  Eg <- seq(-0.02, 0.09, length.out = 500)
  stopifnot(max(abs(sd_at(fr, Eg)^2 -
    (fr$C * Eg^2 - 2 * fr$A * Eg + fr$B) / fr$D)) < 1e-15)
}
stopifnot(abs(sum(f_hist$xmin) - 1) < 1e-9, abs(sum(f_sim$xmin) - 1) < 1e-9)

cat("3. frontier inputs\n")
cat(sprintf("  %-13s %9s %9s %9s %9s %9s %9s\n",
  "matrix", "A", "B", "C", "D", "E min", "sd min"))
cat(sprintf("  %-13s %9.4f %9.5f %9.2f %9.2f %8.3f%% %8.3f%%\n",
  "historical", f_hist$A, f_hist$B, f_hist$C, f_hist$D,
  f_hist$Emin * 100, f_hist$sdmin * 100))
cat(sprintf("  %-13s %9.4f %9.5f %9.2f %9.2f %8.3f%% %8.3f%%\n\n",
  "single index", f_sim$A, f_sim$B, f_sim$C, f_sim$D,
  f_sim$Emin * 100, f_sim$sdmin * 100))

E_cmp <- 0.02
gap <- sd_at(f_sim, E_cmp) - sd_at(f_hist, E_cmp)
cat(sprintf("  at E = %.0f%%: historical %.4f%%, single index %.4f%% (+%.4f pp)\n",

```

```

      E_cmp * 100, sd_at(f_hist, E_cmp) * 100,
      sd_at(f_sim, E_cmp) * 100, gap * 100))

dgap <- function(E) sd_at(f_sim, E) - sd_at(f_hist, E)
xhi <- uniroot(dgap, c(0.02, 0.30), tol = 1e-12)$root
xlo <- uniroot(dgap, c(-0.30, 0.01), tol = 1e-12)$root
stopifnot(dgap(mean(c(xlo, xhi))) > 0, dgap(xhi + 0.01) < 0)

cat(sprintf("  asymptote slope sqrt(D/C): historical %.4f, single index %.4f\n",
            sqrt(f_hist$D / f_hist$C), sqrt(f_sim$D / f_sim$C)))
cat(sprintf("  the curves cross at E = %.2f%% and E = %.2f%%; between them the\n",
            xlo * 100, xhi * 100))
cat(sprintf("  single index frontier is to the right, outside them to the left\n\n"))

real <- function(x) sqrt(as.numeric(t(x) ** S_hist ** x))
stopifnot(real(f_sim$xmin) >= real(f_hist$xmin)) # historical is the true floor

cat("  minimum risk portfolio, scored under the historical matrix:\n")
cat(sprintf("    historical weights    %.4f%%    (the in-sample floor)\n",
            real(f_hist$xmin) * 100))
cat(sprintf("    single index weights %.4f%%    (single index claims %.4f%%,\n",
            real(f_sim$xmin) * 100, f_sim$sdmin * 100))
cat(sprintf("    so it understates its own risk by %.4f pp)\n",
            (real(f_sim$xmin) - f_sim$sdmin) * 100))
cat(sprintf("    but the weights are far less extreme: sum|w| %.2f vs %.2f,\n",
            sum(abs(f_sim$xmin)), sum(abs(f_hist$xmin))))
cat(sprintf("    largest %.3f vs %.3f, %d shorts vs %d\n\n",
            max(f_sim$xmin), max(f_hist$xmin),
            sum(f_sim$xmin < 0), sum(f_hist$xmin < 0)))

# ---- plots -----

surface <- "#fcbcfb"; ink <- "#0b0b0b"; muted <- "#898781"; grid <- "#e1e0d9"
c_hist <- "#2a78d6" # historical covariance matrix (Project 2)
c_sim <- "#eb6834" # single index covariance matrix
c_spx <- "#1baf7a"

open_png <- function(f, h = 4.7) {
  png(f, width = 6.5, height = h, units = "in", res = 200)
  par(bg = surface, mar = c(4.2, 4.6, 3.4, 1.4), family = "sans")
}

ax <- function(side, at, lab) {
  axis(side, at = at, labels = lab, col = "#c3c2b7", col.axis = muted,
       cex.axis = 0.75, lwd = 1, las = if (side == 2) 1 else 0)
}

pt <- function(x, y, col, lab, pos = 4, cex = 0.72, off = 0.6) {
  points(x, y, pch = 21, bg = col, col = surface, lwd = 1.8, cex = 1.5)
  text(x, y, lab, pos = pos, offset = off, cex = cex, font = 2, col = col)
}

means_all <- colMeans(ret); sd_all <- sqrt(diag(cov(ret)))
is_g <- colnames(ret) == "^GSPC"

```

```

draw_both <- function(xr, yr, title, sub) {
  Eg <- seq(yr[1], yr[2], length.out = 1500)
  plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
  abline(h = pretty(yr), v = pretty(xr), col = grid, lwd = 1)
  for (fr in list(list(f_hist, c_hist), list(f_sim, c_sim))) {
    s <- sd_at(fr[[1]], Eg); eff <- Eg >= fr[[1]]$Emin
    lines(s[!eff], Eg[!eff], col = fr[[2]], lwd = 1.2, lty = 3)
    lines(s[ eff], Eg[ eff], col = fr[[2]], lwd = 2.6)
  }
  ax(1, pretty(xr), sprintf("%.0f%%", pretty(xr) * 100))
  ax(2, pretty(yr), sprintf("%.0f%%", pretty(yr) * 100))
  mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
  mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
  mtext(title, 3, line = 1.8, adj = 0, col = ink, cex = 1.02, font = 2)
  mtext(sub, 3, line = 0.5, adj = 0, col = muted, cex = 0.72)
}

open_png("fig_frontiers.png", h = 5.2)
xr <- c(0, max(sd_all) * 1.06)
yr <- c(min(means_all) - 0.009, max(means_all) + 0.007)
draw_both(xr, yr, "Two frontiers: same means, different covariance matrices",
  "Solid: efficient half. Dotted: inefficient half. Grey: the 30 stocks.")
points(sd_all[!is_g], means_all[!is_g], pch = 21, bg = "#d9d8d2",
  col = surface, lwd = 1, cex = 0.8)
points(sd_at(f_hist, xhi), xhi, pch = 4, col = ink, lwd = 2, cex = 1.1)
text(sd_at(f_hist, xhi), xhi, sprintf("curves cross, E = %.2f%%", xhi * 100),
  pos = 4, offset = 0.5, cex = 0.66, font = 2, col = ink)
pt(sd_all[is_g], means_all[is_g], c_spx, "S&P 500", 1)
pt(f_hist$sdmin, f_hist$Emin, c_hist, "historical", 2)
pt(f_sim$sdmin, f_sim$Emin, c_sim, "single index", 4)
dev.off()

open_png("fig_frontiers_zoom.png")
xr2 <- c(0, max(f_sim$sdmin, f_hist$sdmin) * 2.3)
yr2 <- c(min(f_hist$Emin, f_sim$Emin) - 0.014, 0.05)
draw_both(xr2, yr2, "The same two frontiers, near the vertex",
  "Over the range the data occupies, the single index frontier sits to the right.")
segments(sd_at(f_hist, E_cmp), E_cmp, sd_at(f_sim, E_cmp), E_cmp,
  col = ink, lwd = 1.4)
text(sd_at(f_sim, E_cmp), E_cmp, sprintf("+%.2f pp at E = 2%", gap * 100),
  pos = 4, offset = 0.4, cex = 0.68, font = 2, col = ink)
pt(f_hist$sdmin, f_hist$Emin, c_hist, "historical", 2)
pt(f_sim$sdmin, f_sim$Emin, c_sim, "single index", 4)
dev.off()

cat("wrote fig_frontiers.png, fig_frontiers_zoom.png\n")

```